

DATA ANALYST PORTFOLIO PROJECT

Sales & Business Analytics Dashboard

Power BI | Data Analysis | Business Intelligence

"Transforming sales data into actionable business insights."



Power BI



Data Analytics



Business Intelligence



INTRODUCTION

Project Overview

This project uses historical sales and order data to build a centralized Power BI dashboard that consolidates revenue, cost, profitability, and volume metrics into a single, interactive view. Rather than reviewing scattered spreadsheet totals, stakeholders can explore performance across products, regions, and sales channels in real time.



Revenue



Profit



Cost



Units Sold



Product Performance



Regional Performance

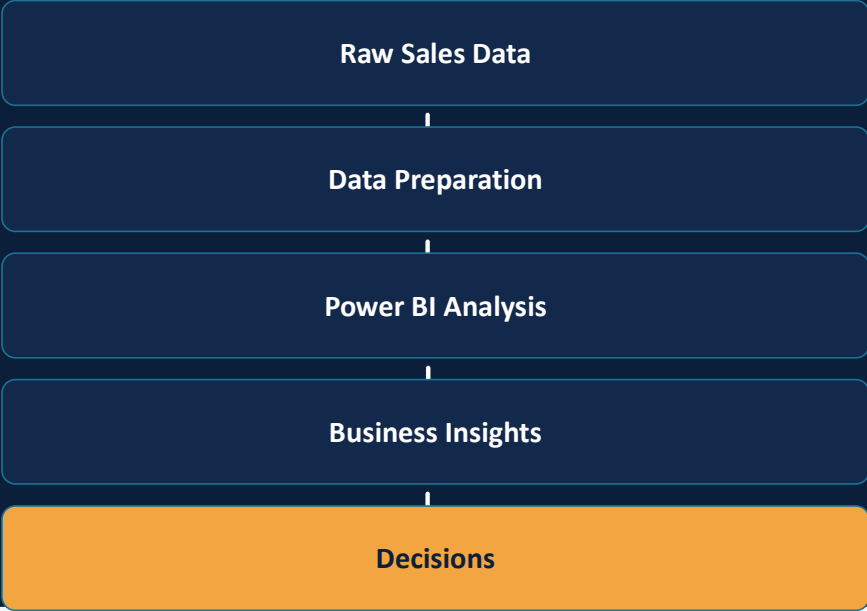


Sales Channel
Performance



Order Trends

HOW POWER BI ADDS VALUE



GOALS

Project Objectives



Monitor Sales & Profitability

Track overall revenue, cost and profit performance in one view.



Analyze Regional Performance

Compare revenue and profit across countries and regions.



Identify Item Type Performance

Surface high- and low-performing item types by profit.



Compare Channels & Priorities

Evaluate sales channels and order priority contribution.



Track Trends Over Time

Observe how cost, revenue and profit move year over year.



Support Decisions

Enable faster, data-driven business decision making.

THE CHALLENGE

Problem Statement

Without a centralized view, raw sales data makes it difficult to:

- ⚠ Quickly identify the most profitable item types
- ⚠ Compare performance across regions and countries
- ⚠ Understand each sales channel's contribution
- ⚠ Track revenue and profit trends over time
- ⚠ Identify areas that need business attention
- ⚠ Make decisions from scattered spreadsheet data

THE SOLUTION

Power BI Dashboard



Centralized KPIs



Interactive Analysis



Faster Decision Making

THE DATA

Dataset Overview



Business Dimensions

Country, Region, Item Type, Rep Name, Sales Channel, Order Priority



Dates

Order Date, Ship Date



Identifiers

Order ID



Financial Metrics

Total Cost, Total Revenue, Total Profit, Unit Cost, Unit Price



Sales Volume

Units Sold

Geography + Products + Sales Channels + Time + Financial Performance

Data Modeling & Preparation

Sales Dataset

Data Cleaning & Transformation

Power BI Data Model

Measures / KPIs (DAX)

Interactive Dashboard

DATA PREPARATION ACTIVITIES

-  Checked for missing values
-  Removed duplicate records where applicable
-  Corrected and standardized data types
-  Formatted Order Date and Ship Date fields
-  Validated numeric financial fields
-  Created calculated measures and KPIs

Built on a single flat sales table using Power BI's native modeling and DAX measures — no external ETL pipeline or multi-table schema was required for this analysis.

THE SOLUTION

Dashboard Overview

The Power BI dashboard brings KPI summaries, trend analysis, and comparative breakdowns together on a single screen, with a Year slicer that lets users filter every visual interactively.



KPI Summary Cards

Order ID count, Total Cost, Total Profit and Total Revenue at a glance.



Profit by Item Type

Ranked bar chart of total profit across all item types.



Revenue, Profit, Cost & Order Trend

Combo line chart comparing four metrics across item types.



Cost Trend by Year

Line chart tracking total cost from 2010 through 2017, with forecast.



Profit by Sales Channel

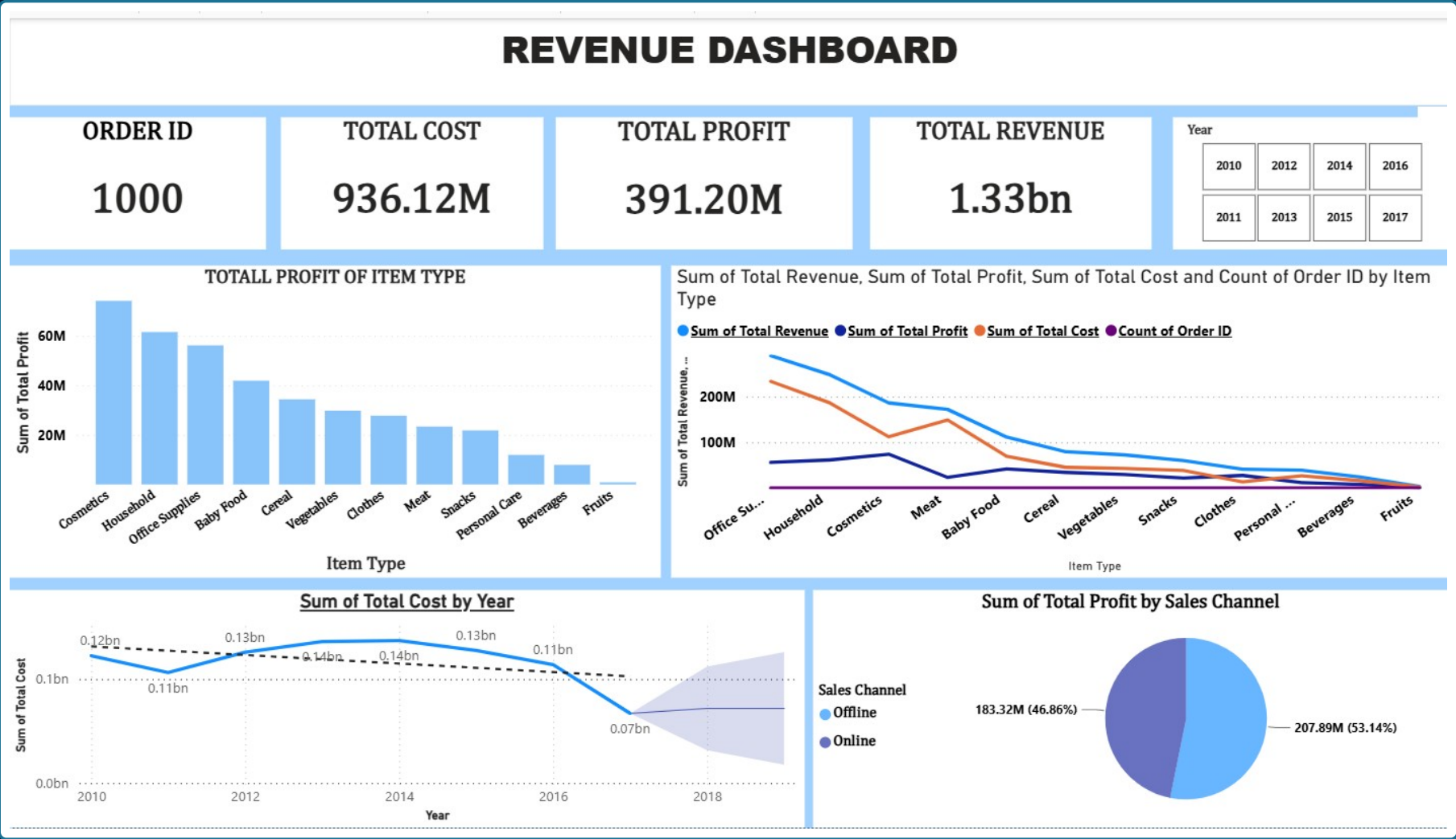
Pie chart splitting total profit between Offline and Online.



Year Filter

Interactive slicer (2010–2017) to filter the full dashboard.

INTERACTIVE POWER BI DASHBOARD



Filters: Year Slicer (2010–2017)

- KPI Summary
- Item Type Profit Ranking
- Revenue / Profit / Cost Trend
- Cost Trend by Year
- Profit by Sales Channel

WHAT THE DATA SHOWS

Key Insights

1.33bn

Total Revenue

391.20M

Total Profit

936.12M

Total Cost

1,000

Orders Analyzed



Cosmetics leads profitability

Cosmetics generates the highest total profit of any item type, followed by Household and Office Supplies — these three categories anchor overall profitability.



Office Supplies drives the top line

Office Supplies records the highest combined revenue, cost and order count among item types, making it the largest contributor to overall business volume.



Offline channel leads, but Online is close

Offline sales contribute 53.14% of total profit (207.89M) versus 46.86% (183.32M) from Online — a fairly balanced channel mix worth monitoring.



Total cost dropped sharply after 2016

Total cost fell from roughly 0.11bn in 2016 to about 0.07bn in 2017, the steepest year-over-year decline in the trend, with a widening forecast range beyond it.



Fruits, Beverages & Personal Care lag

These item types sit at the bottom of the profit ranking, indicating limited contribution compared to top performers like Cosmetics and Household.

Business Recommendations



Double down on top performers

Prioritize marketing and inventory investment in Cosmetics, Household and Office Supplies given their proven profit contribution.



Investigate low-profit categories

Review pricing, cost structure and demand for Fruits, Beverages and Personal Care to identify improvement opportunities.



Refine sales channel strategy

With Offline at 53% and Online at 47% of profit, evaluate whether targeted investment can grow the Online channel's share.



Monitor cost and margin closely

Track the cost trend that dropped sharply after 2016 to confirm it reflects efficiency gains rather than a data or volume issue.



Establish regular Power BI reporting

Use the dashboard as a recurring reporting cadence so performance shifts are caught early and decisions stay data-driven.

Tools & Skills



Power BI

- ✓ Dashboard Development
- ✓ Data Visualization
- ✓ KPI Development
- ✓ Interactive Filters / Slicers



DAX

- ✓ Measures
- ✓ Calculated Metrics
- ✓ Business KPIs



Data Analysis

- ✓ Trend Analysis
- ✓ Sales Analysis
- ✓ Profitability Analysis
- ✓ Business Insights

CONCLUSION

Sales Data, Transformed Into Insight

This project demonstrates how Power BI can transform raw sales data into a centralized analytical solution — turning scattered numbers into a clear, explorable story for the business.

- ✓ Improved visibility into overall sales performance
- ✓ Better understanding of revenue and profitability drivers
- ✓ Easier comparison across products, regions and channels
- ✓ Interactive, self-serve business analysis
- ✓ Stronger support for data-driven decisions

“The dashboard transforms raw sales data into clear, actionable business insights.”

Thank You